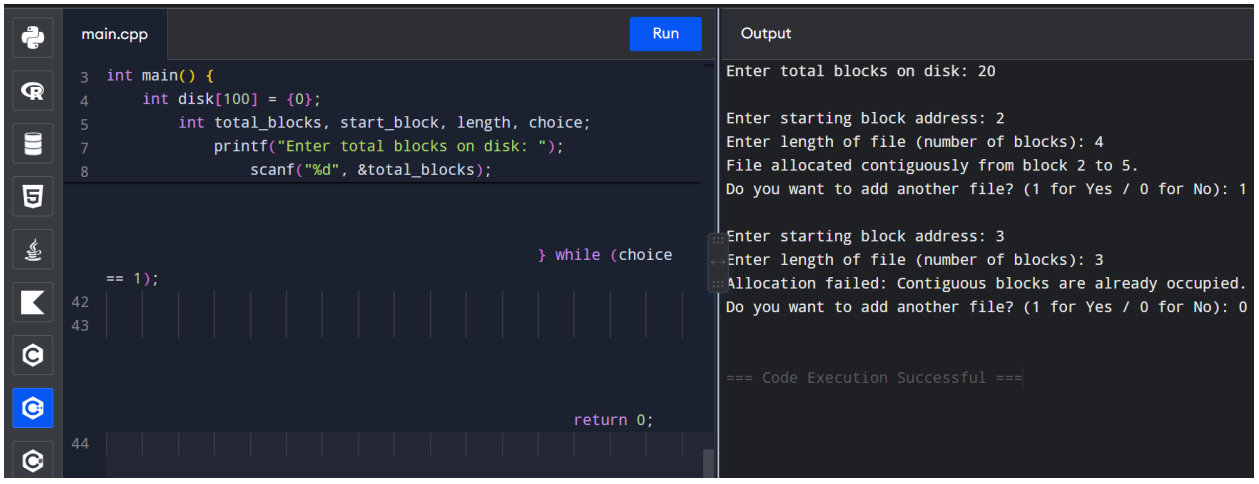
EXPERIMENT 28:

28. Consider a file system where the records of the file are stored one after another both physically and logically. A record of the file can only be accessed by reading all the previous records. Design a C program to simulate the file allocation strategy.

AIM:
To design a C program to simulate the Sequential File Allocation strategy where records are stored contiguously.



PSEUDO CODE :

BEGIN

PRINT "Enter total blocks on disk:"

READ total_blocks

DECLARE disk array of size total_blocks INITIALIZED to 0

LOOP

PRINT "Enter file starting block and length (number of blocks):"

READ start_block, length

SET is_allocated = 1

```
        // Check if start bounds and contiguous blocks are free
        IF start_block + length > total_blocks THEN
            PRINT "Error: Out of disk bounds!"
            SET is_allocated = 0
        ELSE
            FOR i FROM start_block TO (start_block + length - 1) DO
                IF disk[i] == 1 THEN
                    SET is_allocated = 0
                    BREAK
                END IF
            END FOR
        END IF

        IF is_allocated == 1 THEN
            FOR i FROM start_block TO (start_block + length - 1) DO
                SET disk[i] = 1
            END FOR
            PRINT "File successfully allocated from block " + start_block + " to " +
(start_block + length - 1)
        ELSE
            PRINT "Allocation failed: Blocks requested are not free or invalid."
        END IF

        PRINT "Do you want to allocate another file? (1-Yes, 0-No):"
        READ choice
    UNTIL choice == 0
END
```

CODE:

```
#include <stdio.h>

int main() {

    int disk[100] = {0};

    int total_blocks,
start_block, length,
choice;



    printf("Enter total
blocks on disk: ");

    scanf("%d",
&total_blocks);



    do {

        printf("\nEnter
starting block address:
");

        scanf("%d",
&start_block);

        printf("Enter length
```

```
of file (number of
blocks): ");
        scanf("%d",
&length);

        int is_allocated = 1;

        if (start_block +
length > total_blocks ||
start_block < 0) {
            printf("Error:
Request exceeds total
disk allocation
bounds.\n");
            is_allocated = 0;
        } else {
            for (int i =
start_block; i <
start_block + length;
i++) {
```

```
            if (disk[i] ==
1) {
                is_allocated
= 0;
                break;
            }
        }
    }

    if (is_allocated) {
        for (int i =
start_block; i <
start_block + length;
i++) {
            disk[i] = 1;
        }
        printf("File
allocated contiguously
from block %d to
%d.\n", start_block,
```

```
start_block + length -
1);

        } else if
(start_block + length <=
total_blocks &&
start_block >= 0) {


printf("Allocation
failed: Contiguous
blocks are already
occupied.\n");

        }



        printf("Do you
want to add another file?
(1 for Yes / 0 for No):
");

        scanf("%d",
&choice);

    } while (choice == 1);
```

```
    return 0;

}
```